GGSO: Graph-Guided Surge Optimization for Network-Wide Ride-Hailing Pricing

Abstract: This project develops GGSO, an automated pricing system that generates network-wide surge adjustments for ride-hailing operations. Every five minutes, the DuoCast model forecasts booking demand and available drivers across H3 cells. It uses spatial adjacency and observed driver-mobility networks to capture relationships between areas. Cell-level response models then estimate how candidate prices may affect bookings and fulfillment, while a mixed-integer linear program selects a coordinated network pricing policy under business and operational constraints. Validated policies are applied automatically, while a monitoring dashboard provides execution status and a complete audit trail.

Keywords: Ride-hailing, surge pricing, spatiotemporal GNN, demand elasticity, conservative optimization, MILP, automated pricing.

1. Introduction and Business Problem

Ride-hailing demand and driver availability change quickly across locations and five-minute intervals. One H3 cell may face a shortage while a nearby cell has excess supply. These cells are not independent because drivers move across cell boundaries and market conditions can extend beyond a single area [1, 2].

The current surge system assigns a multiplier to each H3 cell using only local information. This approach captures the market conditions within each cell, but it does not account for the spatial relationships among cells in a ride-hailing network. Similar local demand-supply conditions may therefore have different implications depending on nearby supply and observed driver movements.

GGSO builds on the local approach by incorporating information from neighboring cells and the observed driver-mobility network. H3 adjacency represents spatial neighborhood relationships, while directed mobility links describe how drivers move between cells. This network context supports coordinated surge decisions that reflect market conditions beyond individual cells.

GGSO evaluates surge multipliers jointly across the network rather than treating each cell in isolation. Its business objective is to improve network-level gross merchandise value (GMV) while maintaining acceptable booking rates and valid pricing. Every five minutes, the system automatically applies a complete network pricing policy that passes all validation checks. If validation fails, the current effective prices remain unchanged.

2. Proposed GGSO Framework

GGSO consists of three core components. **DuoCast** forecasts booking demand and available drivers for every H3 cell using local, spatial, and driver-mobility information. **Response models** estimate how candidate surge multipliers may change bookings and fulfillment in each cell. **Network optimization** uses a mixed-integer linear program (MILP) to select one candidate per cell while enforcing network and pricing constraints.

At each five-minute decision cutoff, GGSO freezes a complete network snapshot containing historical marketplace observations, spatial and mobility graphs, current effective surge multipliers, and the operational context required by the response models. DuoCast runs once on this snapshot to forecast demand and available supply for every H3 cell.

The response layer combines these forecasts with the fitted response parameters to evaluate the eligible pricing candidates. The MILP then selects one candidate for each cell as a coordinated network policy. An independent validator recomputes network coverage, hard constraints, and objective components before the policy is applied automatically. If validation fails, the current effective prices remain unchanged. Figure 1 shows this end-to-end flow.

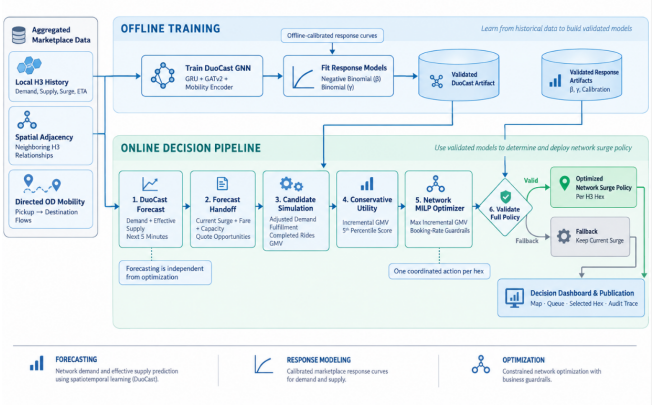


Figure 1: End-to-end GGSO architecture: point-in-time data, forecast, candidate simulation, MILP, validation, automated execution, and monitoring dashboard.

3. DuoCast Model

3.1 Spatiotemporal Representation

DuoCast is a temporal, spatial, and directed-mobility GNN. For each H3 cell i, a shared gated recurrent unit (GRU) encodes historical marketplace observations. Two graph-attention branches aggregate context from spatially adjacent cells and directed, observed physical driver movements, respectively. A gated residual fusion combines the temporal, spatial, and mobility representations before two separate demand and supply prediction heads [3, 4, 1]. The forecasting function is written as

$$(\widehat{D}_i, \widehat{S}_i) = f_\theta(X_{t-T+1:t}, G_{\mathrm{sp}}, G_{\mathrm{od},t}), \tag{1}$$

where X is the history available at cutoff t; T is the number of observed snapshots; G_{sp} is the H3 adjacency graph; $G_{\mathrm{od},t}$ is the directed historical driver-mobility graph; and θ is the set of trainable parameters. The model produces two outputs for each H3 cell: $\widehat{D}_i$, the forecast booking demand in $(t, t+5]$, and $\widehat{S}_i$, the forecast available driver supply at $t+5$. Driver supply S is distinct from service capacity K.

Spatial and mobility edges encode different relationships. H3 adjacency represents the neighboring-cell structure, whereas the mobility graph summarizes directed historical driver movements between cells; it is not the same as passenger trip demand and does not predict future repositioning caused by a new surge price. A model may improve forecasts under the historical policy and still become unreliable outside the market conditions represented by its data.

3.2 Multi-Task Training Objective

Demand and supply are trained jointly with equal contributions to the loss. Both nonnegative targets are transformed using $\log(1+x)$ to reduce scale differences between sparse and high-volume cells. The training objective is

$$\mathcal{L}_{\text{fc}} = \frac{1}{2N}\sum_{i=1}^{N}\Big[\rho_\delta(\widehat{\ell}_i^{D} - \log(1+D_i)) + \rho_\delta(\widehat{\ell}_i^{S} - \log(1+S_i))\Big], \tag{2}$$

where N is the number of H3 cells; D_i is the observed demand over the next five-minute interval; S_i is the observed supply at the target time; $\widehat{\ell}_i^{D}$ and $\widehat{\ell}_i^{S}$ are the corresponding log-space predictions; and ρ_δ is the Huber penalty with threshold δ. The factor $1/(2N)$ averages the loss across cells and the two targets, so neither task receives an additional weight. Predictions are decoded as $\widehat{y} = \max\{\exp(\widehat{\ell}) - 1, 0\}$. The Huber penalty is defined in Appendix B.

4. Marketplace Response and Optimization

4.1 Cell-Level Response Models

GGSO fits two generalized linear models (GLMs) offline and independently for each eligible H3 cell using historical data. The demand GLM uses applied surge, booking counts, and demand context known before the pricing decision. The fulfillment GLM uses matched booking and completion cohorts, effective service capacity, and its own pre-decision context. At decision time, GGSO loads the fitted model parameters rather than refitting either model for each candidate.

Demand response GLM. Booking demand is modeled with a Negative Binomial GLM:

$$\begin{aligned} D_{i,t} &\sim \text{NegBin}(\mu_{i,t}, \kappa_i), \\ \log \mu_{i,t} &= \alpha_i + w_i^\top c_{i,t}^{D} - \beta_i \log s_{i,t}, \end{aligned} \tag{3}$$

Here $D_{i,t}$ is the observed booking count, $\mu_{i,t}$ is its expected value, and κ_i is the dispersion parameter. The vector $c_{i,t}^{D}$ contains demand context known before the pricing decision, while $s_{i,t} > 0$ is the applied surge multiplier. The parameters α_i and w_i capture the cell baseline and context effects. The coefficient $0 \le \beta_i \le 4$ controls the estimated demand response to surge: a larger β_i implies a stronger decrease in expected bookings as surge increases. During candidate simulation, β_i adjusts the DuoCast demand forecast; the fitted mean $\mu_{i,t}$ does not replace that forecast.

Fulfillment response GLM. Among the requests in the same matured booking cohort, completed rides follow a Binomial GLM:

$$\begin{aligned} C_{i,t} &\sim \text{Binomial}(D_{i,t}, \pi_{i,t}), \\ \text{logit}(\pi_{i,t}) &= b_i + v_i^\top c_{i,t}^{F} + \gamma_i \log\frac{K_{i,t}+\varepsilon}{D_{i,t}+\varepsilon}. \end{aligned} \tag{4}$$

Here $C_{i,t}$ counts completed rides from the $D_{i,t}$ booking cohort, and $\pi_{i,t}$ is the probability that a booking is fulfilled before the explicit capacity cap is applied. The estimate depends on fulfillment context $c_{i,t}^{F}$ and on the ratio between service capacity $K_{i,t}$ and booking demand. The parameters b_i and v_i capture the cell baseline and context effects, $\varepsilon > 0$ stabilizes the ratio, and $0 \le \gamma_i \le 1$ controls its influence on fulfillment. The model output is the fulfillment probability $\pi_{i,t}$, which becomes the fulfillment response used in candidate simulation.

The models remain separate because booking behavior and operational fulfillment are different mechanisms. Surge enters the demand model directly, whereas the fulfillment model responds to capacity relative to demand. Their fitted coefficients are combined during candidate simulation, with the final completion estimate capped by available capacity. Both models require separate support and calibration checks. Because they are estimated from observational data, their coefficients are not treated as established causal price effects. Appendix C gives the likelihood objectives and fitting checks.

4.2 Candidate Simulation and Conservative Scoring

For each cell, GGSO applies an adjustment $a \in \mathcal{A} = \{-0.3, -0.2, \ldots, 0.2, 0.3\}$ to the current surge s_i^0, giving $s_i(a) = s_i^0 + a$. Candidates outside the configured pricing bounds or historical support are rejected. For each eligible candidate, GGSO estimates demand, completed rides, and modeled GMV as

$$\widehat{D}_i(a) = \widehat{D}_i\left(\frac{s_i(a)}{s_i^0}\right)^{-\beta_i}, \tag{5}$$

$$\begin{aligned} \widehat{C}_i(a) &= \min\{\widehat{D}_i(a) FR_i(a), K_i\}, \\ U_i(a) &= \widehat{C}_i(a)\, p_i\, s_i(a), \end{aligned} \tag{6}$$

where $FR_i(a)$ is the fitted fulfillment response and p_i is the reference fare before surge. Thus, $U_i(a)$ is the modeled GMV for candidate a, not its improvement over the current policy.

The GLM coefficients β_i and γ_i are uncertain because they are estimated from historical data. The response artifact therefore stores B paired parameter scenarios $\vartheta_i^{(b)} = (\beta_i^{(b)}, \gamma_i^{(b)})$. Under each scenario, GGSO recomputes both the candidate and the reference action using the same parameter pair:

$$\begin{aligned} \Delta U_i^{(b)}(a) &= U_i(a; \vartheta_i^{(b)}) - U_i(0; \vartheta_i^{(b)}), \\ \widetilde{U}_i(a) &= \text{Quantile}_{0.05}\left(\left\{\Delta U_i^{(b)}(a)\right\}_{b=1}^{B}\right). \end{aligned} \tag{7}$$

The fifth percentile represents lower-performing parameter scenarios and is used as a conservative score to reduce the risk of overconfident decisions.

All candidates share one DuoCast forecast under the current policy. The simulator changes the response calculations for each action; it does not rerun DuoCast or predict driver redistribution. Its validated candidate table is the input to the network optimizer.

4.3 Joint Network MILP

Let $z_{i,a} \in \{0,1\}$ indicate whether cell i selects adjustment a, with exactly one permitted candidate selected for each cell, and let s_i denote the resulting surge multiplier. The network optimizer solves

$$\begin{aligned} \max_{z} \quad & \sum_{i,a} z_{i,a}\widetilde{U}_i(a) - \lambda_t \sum_{i,a} z_{i,a}|a| \\ & - \lambda_g \sum_{(i,j)\in\mathcal{E}} w_{ij}|s_i - s_j|, \end{aligned} \tag{8}$$

The first term rewards conservative incremental GMV. The second discourages large adjustments from the current surge, while the third discourages large surge differences between neighboring cells. The coefficients λ_t and λ_g control the two

penalties, and w_{ij} is the weight of edge (i, j). The solver linearizes $|s_i - s_j|$ internally as described in Appendix E. Two principal hard constraints are

$$BR_{\min} Q \le \sum_{i,a} z_{i,a} \widehat{D}_i(a) \le BR_{\max} Q,$$
$$\sum_{i,a} z_{i,a} \Delta U_i(a) \ge 0, \qquad Q = \sum_i Q_i > 0, \tag{9}$$

where Q_i is the compatible number of quote opportunities, and $\Delta U_i(a) = U_i(a) - U_i(0)$ is nominal modeled incremental GMV. The no-loss constraint uses nominal modeled incremental GMV, while the objective uses the conservative lower-tail score. It prevents a negative nominal network total but does not guarantee realized GMV. HiGHS (High Performance Optimization Software) solves the MILP, and an independent validator checks network coverage and all hard constraints before the selected policy is applied. Further details are in Appendix E.

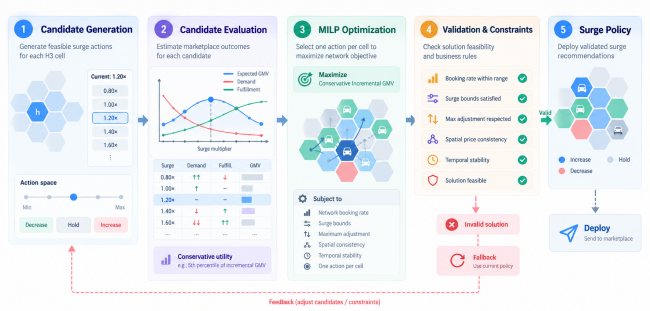


Figure 2: Candidate-to-decision flow: supported price grid, response simulation, paired uncertainty, joint MILP, and independent validity gate.

5. Validation and Evidence

The evaluation assesses both the technical performance of GGSO and its impact on business outcomes.

5.1 Forecasting Performance

DuoCast is evaluated on a chronologically held-out test set using raw-space and log-space forecasting metrics. LightGBM [5], GRU-only, Spatial GATv2, and Mobility GATv2 use the same data split, forecast horizon, node set, and target definitions.

Table 1: Forecasting performance on the held-out test set.

Model	D-MAE ↓	S-MAE ↓	Log D-MAE ↓	Log S-MAE ↓	Log D/S MAE ↓	Log D/S RMSE ↓
LightGBM baseline	74.2329	5.8163	0.5149	0.1639	0.4587	0.5794
GRU-only	10.0524	4.6252	0.0780	0.1317	0.1489	0.1898
Spatial GATv2	9.8521	4.6257	0.0768	0.1318	0.1484	0.1890
Mobility GATv2	9.8627	4.6268	0.0769	0.1318	0.1484	0.1890
DuoCast	**9.7866**	**4.6237**	**0.0765**	**0.1317**	**0.1482**	**0.1888**

Table 1 compares DuoCast with the baseline and ablation models on the held-out test set. DuoCast achieves the lowest demand MAE, supply MAE, log-demand MAE, log demand-to-supply MAE, and log demand-to-supply RMSE, while matching the GRU-only result for log-supply MAE.

Compared with the GRU-only baseline, DuoCast reduces demand MAE by 2.64% and supply MAE by 0.03%. The MAE and RMSE of the log demand-to-supply ratio are reduced by 0.47% and 0.53%, respectively.

Both Spatial GATv2 and Mobility GATv2 outperform the GRU-only baseline on demand forecasting and demand-to-supply ratio estimation, indicating that spatial adjacency and directed driver mobility provide complementary predictive information.

5.2 Response Model Performance

The demand and fulfillment GLMs are evaluated independently on chronologically held-out observations. For the demand model, prediction error is summarized using MAE and RMSE. For the fulfillment model, fulfillment-rate MAE measures the difference between predicted and observed fulfillment rates. The evaluation also reports the fitted response coefficients, their uncertainty, and the proportion of H3 cells with validated response models.

Table 2: Negative Binomial demand GLM performance

Metric	Value
Demand MAE ↓	6.85
Demand RMSE ↓	10.42
Median β	1.20
Median 95% CI Width (β)	0.28
Eligible H3 Cells (%) ↑	92.5%

Table 3: Binomial fulfillment GLM performance .

Metric	Value
FR MAE ↓	0.028
Median γ	0.65
Median 95% CI Width (γ)	0.18
Eligible H3 Cells (%) ↑	94.2%

The median values of β and γ summarize the estimated demand and fulfillment responses across eligible H3 cells. Their interval widths indicate the uncertainty of these estimates, while eligible-cell coverage reports the proportion of required cells with successfully fitted and validated response models.

These metrics are evaluated against outcomes observed under historically applied surge multipliers. They measure predictive reliability but do not establish causal responses to previously unobserved price changes.

5.3 Optimization and Safety Validation

For small test networks, the MILP solution is compared with exhaustive enumeration of all feasible candidate combinations. This comparison verifies the objective, candidate selection, Booking Rate bounds, nominal modeled GMV no-loss constraint, and spatial penalty.

For each full-network decision, an independent validator recomputes candidate eligibility, network coverage, hard constraints, and objective components from the selected candidates. Any critical validation failure marks the decision as invalid, and no new pricing policy is applied.

5.4 Real-World Business Evaluation

A prospective switchback experiment evaluates GGSO over a sufficiently large operational region containing multiple connected H3 cells. The entire region alternates between the GGSO policy and the current policy across pre-specified randomized time blocks [6]. Total realized GMV across the experimental region is measured for each block as the primary outcome, and the estimated treatment effect is reported with confidence intervals. Until this experiment is completed, GGSO's business impact remains unverified.

6. Automated Pricing and Monitoring

Every five minutes, GGSO uses the latest marketplace data to forecast demand and supply, evaluate surge candidates, and select a network-wide pricing policy. A valid policy is applied automatically. If validation fails, the current effective prices remain unchanged.

The monitoring dashboard displays the H3 map, observed and forecast demand and supply, current, recommended, and applied surge multipliers, decision status, validation errors, and execution history. It updates after each pricing cycle and retains the decision and version identity of each result.

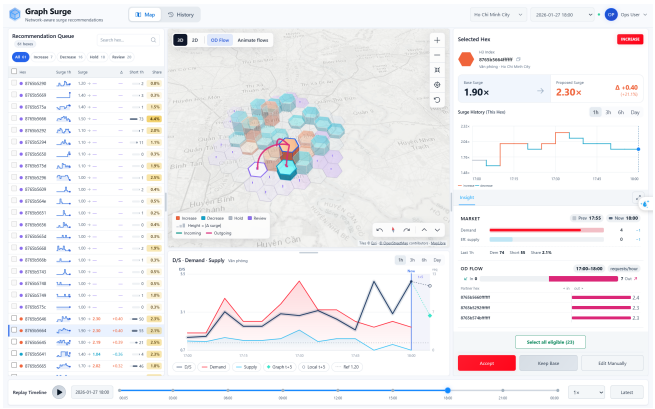


Figure 3: GGSO monitoring dashboard with marketplace forecasts, surge recommendations, execution status, and audit history.

The backend checks price bounds, precision, reference versions, and idempotency before applying a policy, then records the execution receipt. Each new cycle uses the latest confirmed effective surge multipliers as its reference policy.

7. Conclusion and Next Steps

The current surge system determines a multiplier for each H3 cell from its local market conditions. GGSO extends this approach by incorporating spatial adjacency and directed driver mobility across cells. DuoCast forecasts demand and available supply, the response models estimate how each cell may react to candidate multipliers, and the MILP selects a coordinated network pricing policy subject to business and pricing constraints.

GGSO is designed to apply a validated network-wide pricing policy automatically every five minutes, while the monitoring dashboard provides visibility into forecasts, pricing decisions, validation results, and execution history. The next steps are to validate the forecasting and response models, verify the optimization pipeline in shadow mode, and evaluate the complete system through a real-world switchback experiment. Until that experiment is completed, GGSO's realized business impact remains unverified. Its business value should ultimately be judged by its causal effect on realized network GMV.

APPENDIX

A. Notation and Data Definitions

A.1 Decision Setting

Let $\mathcal{V}$ be the set of H3 resolution-7 cells and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ the pricing-adjacency edges. At decision time t, GGSO uses the current surge vector s^0 and data available up to that time. The forecast horizon is $H = 5$ minutes, and T denotes the configurable number of historical five-minute snapshots used by DuoCast.

Table 4: Core notation and compatible units.

Symbol	Definition
$\mathcal{V}, \mathcal{E}$	Set of H3 cells and pricing-adjacency edges.
i, j	H3 cell indices.
t, T, H	Decision cutoff, history length, forecast horizon (minutes).
$D_i, \widehat{D}_i$	Observed and predicted confirmed valid booking requests per five-minute cohort.
$S_i, \widehat{S}_i$	Observed and predicted online, idle, dispatch-eligible drivers (stock at timestamp).
K_i	Effective service capacity, measured in rides per five-minute interval; not driver stock.
Q_i, Q	Compatible quote opportunities per cell and network total, with $Q = \sum_i Q_i$.
C_i, FR_i	Cohort-completed rides and pre-cap fulfillment probability.
$s_i^0, s_i(a)$	Effective reference surge and candidate surge multiplier.
$a, \mathcal{A}_i$	Additive surge adjustment and eligible actions for cell i.
p_i, U_i	Reference fare before surge and nominal modeled GMV.
$\Delta U_i, \widetilde{U}_i$	Nominal modeled GMV increment and conservative incremental score.
$G_{\mathrm{sp}}, G_{\mathrm{od}}$	Spatial adjacency and directed observed-driver-mobility graphs.
$z_{i,a}, d_{ij}$	MILP binary choice and auxiliary variable for $\lvert s_i - s_j \rvert$.
β_i, γ_i	Fitted price sensitivity and fulfillment response coefficients.

A.2 Business Data Definitions

Demand: $D_{i,t}$ is the number of confirmed valid booking requests created in cell i during $(t-5, t]$. The forecast $\widehat{D}_i$ refers to the interval $(t, t+5]$.

Supply: $S_{i,t}$ is the number of online, idle, and dispatch-eligible drivers in cell i at time t. The forecast $\widehat{S}_i$ is the corresponding driver stock at $t+5$.

Fulfillment: $C_{i,t}$ is the number of completed rides from the booking cohort represented by $D_{i,t}$, measured over a defined maturity horizon, with $0 \le C_{i,t} \le D_{i,t}$.

Quote opportunities: $Q_{i,t}$ is the number of compatible quote opportunities for the same cell, time window, and booking cohort.

All input features, prices, graph observations, and context variables must be available by cutoff t and aligned to the same decision interval.

B. Forecasting Network and Training Mathematics

B.1 Temporal, Spatial, and Mobility Encoders

Let $x_{i,k} \in \mathbb{R}^F$ be the observed feature vector for cell i at historical step k. The shared temporal encoder computes

$$h_{i,k} = \mathrm{GRU}(x_{i,k}, h_{i,k-1}), \qquad \ell_i = \mathrm{LayerNorm}(h_{i,T}). \tag{10}$$

The spatial relation uses H3 neighborhood edges; the mobility relation uses physical driver transitions directed from origin to destination and available before cutoff. Each relation $r \in \{\mathrm{sp}, \mathrm{od}\}$ has independent attention parameters. For observed edge attributes $q_{ji}^{(r)}$, one relation-aware GATv2 formulation is

$$\begin{aligned} e_{ji}^{(r)} &= a_r^\top \mathrm{LeakyReLU}\big(W_s^{(r)} \ell_j + W_t^{(r)} \ell_i \\ &\qquad + W_e^{(r)} q_{ji}^{(r)}\big), \\ \alpha_{ji}^{(r)} &= \frac{\exp(e_{ji}^{(r)})}{\sum_{k \in \mathcal{N}_r(i)} \exp(e_{ki}^{(r)})}, \\ m_i^{(r)} &= \sum_{j \in \mathcal{N}_r(i)} \alpha_{ji}^{(r)} W_v^{(r)} \ell_j, \end{aligned} \tag{11}$$

where $\mathcal{N}_r(i)$ is the observed incoming-neighbor set under relation r, $e_{ji}^{(r)}$ is an attention logit, $\alpha_{ji}^{(r)}$ is its normalized coefficient, and $m_i^{(r)}$ is the aggregated relation-specific message. Separate parameters allow the two branches to learn different meanings for spatial proximity and observed driver movement.

B.2 Gated Fusion and Prediction Heads

The temporal representation and the messages from the two graph branches are combined through gated residual fusion:

$$\begin{aligned} c_i^{\mathrm{net}} &= W_c[m_i^{(\mathrm{sp})} \| m_i^{(\mathrm{od})}], \\ \omega_i &= \mathrm{sigmoid}\big(W_g[\ell_i \| m_i^{(\mathrm{sp})} \| m_i^{(\mathrm{od})}]\big), \\ z_i &= \mathrm{LayerNorm}(\ell_i + \omega_i \odot c_i^{\mathrm{net}}), \end{aligned} \tag{12}$$

where $\|$ denotes concatenation, $\odot$ denotes element-wise multiplication, and ω_i is a learned gate. Two separate heads output $\widehat{\ell}_i^{D} = f_D(z_i)$ and $\widehat{\ell}_i^{S} = f_S(z_i)$, subsequently decoded into non-negative demand and supply forecasts. The gate controls how much network context is added to the temporal representation for each cell.

B.3 Log Transformation and Huber Loss

Demand and supply targets are transformed using $\widetilde{y} = \log(1+y)$ before training. This transformation is defined for zero-valued observations and compresses the difference between low-volume and high-volume H3 cells.

The prediction error is measured in log space using the Huber loss:

$$\rho_\delta(e) = \begin{cases} \frac{1}{2}e^2, & |e| \le \delta, \\ \delta\left(|e| - \frac{1}{2}\delta\right), & |e| > \delta. \end{cases} \tag{13}$$

where e is the difference between the predicted and observed log-transformed target, and δ controls the transition between the two regions. The loss is quadratic for small errors and linear for large errors, reducing the influence of unusually large demand or supply observations.

C. Marketplace Response Estimation and Capacity

C.1 Demand Negative Binomial Likelihood

The expected booking demand in Eq. (3) is $\mu_{i,t} = \exp(\alpha_i + w_i^\top c_{i,t}^D - \beta_i \log s_{i,t})$. The constrained maximum-likelihood training objective is

$$\mathcal{L}_{D,i} = -\sum_{t \in \mathcal{T}_i^{\mathrm{train}}} \log P_{\mathrm{NB}}(D_{i,t} \mid \mu_{i,t}, \kappa_i), \quad 0 \le \beta_i \le 4. \tag{14}$$

The parameter tuple includes intercept α_i, ordered context coefficients w_i, elasticity β_i, and dispersion κ_i. The Negative Binomial mean–dispersion convention must remain consistent

through fitting and evaluation because alternative parameterizations assign different meanings to κ_i. Observed surge variation and sufficient per-cell data are prerequisites for meaningful estimation.

C.2 Cohort-Level Binomial Likelihood

For Eq. (4), the conditional fulfillment likelihood gives the following negative-log-likelihood objective, with the parameter-independent binomial coefficient omitted:

$$\mathcal{L}_{F,i} = -\sum_{t\in\mathcal{T}_i^{\text{train}}} \big[C_{i,t}\log\pi_{i,t} + (D_{i,t}-C_{i,t})\log(1-\pi_{i,t})\big], \tag{15}$$

subject to $0 \le \gamma_i \le 1$ and coherent cohort counting. When $D = 0, C = 0$, the interval has no positive binomial trials; it cannot be treated as evidence for 100% fulfillment. The coefficient γ_i does not define the response by itself; the baseline b_i, context effects v_i, and capacity-to-demand ratio are also required.

C.3 Available Service Capacity

If a direct, validated capacity estimate is unavailable, the proposed approximation is

$$K_i = \widehat{S}_i\, r_i \frac{H}{60}\min\left\{1, \frac{H}{\max(ETA_i, H)}\right\}, \quad H = 5, \tag{16}$$

where r_i is the assumed average rides per driver per hour and ETA_i is a pickup-time estimate in minutes. Thus, K_i is measured in rides per five minutes, while $\widehat{S}_i$ is measured in drivers. The factor $H/60$ converts the hourly service rate to the forecast horizon, and the final factor reduces capacity when pickup time exceeds that horizon. Equation (16) is a heuristic capacity conversion, not a third trained prediction head. Unit consistency, zero-capacity handling, and calibration require separate validation.

C.4 Interpretation and Causal Limitations

Because the response models are fitted on observational data, the estimated β_i and γ_i are interpreted as predictive associations rather than causal effects. Response parameters are estimated independently for every required H3 cell.

D. Candidate Evaluation and Conservative Scoring

D.1 Candidate Evaluation

For each H3 cell, GGSO evaluates the fixed action grid

$$\mathcal{A} = \{-0.3, -0.2, -0.1, 0, 0.1, 0.2, 0.3\}, \qquad s_i(a) = s_i^0 + a. \tag{17}$$

A candidate is eligible when its multiplier is positive, remains within the configured surge and adjustment bounds, and is supported by historical prices.

For each eligible action, expected demand is

$$\widehat{D}_i(a) = \widehat{D}_i\left(\frac{s_i(a)}{s_i^0}\right)^{-\beta_i}. \tag{18}$$

The baseline fulfillment logit and candidate fulfillment probability are

$$\begin{aligned}\eta_i^0 &= b_i + v_i^\top c_i^F + \gamma_i \log\frac{K_i+\varepsilon}{\widehat{D}_i+\varepsilon},\\ FR_i(a) &= \text{sigmoid}\left(\eta_i^0 + \gamma_i\log\frac{\widehat{D}_i+\varepsilon}{\widehat{D}_i(a)+\varepsilon}\right).\end{aligned} \tag{19}$$

Expected completed rides and GMV are then

$$\begin{aligned}\widehat{C}_i(a) &= \min\{\widehat{D}_i(a)FR_i(a), K_i\},\\ U_i(a) &= \widehat{C}_i(a)p_i s_i(a).\end{aligned} \tag{20}$$

Capacity, fulfillment context, and pre-surge fare remain fixed across all candidates of the same cell and decision interval.

D.2 Conservative Scoring

Let $\vartheta_i^{(b)} = (\beta_i^{(b)}, \gamma_i^{(b)})$ denote a response-parameter scenario. The incremental GMV of action a under scenario b is

$$\Delta U_i^{(b)}(a) = U_i(a;\vartheta_i^{(b)}) - U_i(0;\vartheta_i^{(b)}). \tag{21}$$

The conservative score is the empirical fifth percentile of the paired incremental values:

$$\widetilde{U}_i(a) = \text{Quantile}_{0.05}\left(\{\Delta U_i^{(b)}(a)\}_b\right). \tag{22}$$

The candidate and baseline are recomputed using the same parameter scenario, so the score measures their paired difference under response uncertainty. For the zero action, both the incremental GMV and conservative score are zero.

E. MILP Formulation and Validation

E.1 Decision Variables and Objective

For each cell $i \in \mathcal{V}$ and eligible candidate $a \in \mathcal{A}_i$, let $z_{i,a}$ indicate whether the candidate is selected. Exactly one candidate is selected for each cell:

$$\begin{aligned}z_{i,a} \in \{0,1\}, \qquad \sum_{a\in\mathcal{A}_i} z_{i,a} = 1,\\ s_i = \sum_{a\in\mathcal{A}_i} z_{i,a}s_i(a).\end{aligned} \tag{23}$$

For each pricing edge (i, j), the auxiliary variable d_{ij} linearizes the absolute difference between the selected surge multipliers:

$$d_{ij} \ge 0, \qquad d_{ij} \ge s_i - s_j, \qquad d_{ij} \ge s_j - s_i. \tag{24}$$

The objective maximizes conservative incremental utility while penalizing large changes from the current price and large price differences between neighboring cells, as defined in Eq. (8).

E.2 Hard Constraints

The predicted network Booking Rate is constrained by

$$BR_{\min}Q \le \sum_{i\in\mathcal{V}}\sum_{a\in\mathcal{A}_i} z_{i,a}\widehat{D}_i(a) \le BR_{\max}Q, \qquad Q = \sum_i Q_i > 0. \tag{25}$$

The selected policy must also satisfy the nominal modeled GMV no-loss constraint:

$$\sum_{i\in\mathcal{V}}\sum_{a\in\mathcal{A}_i} z_{i,a}\left[U_i(a) - U_i(0)\right] \ge 0. \tag{26}$$

The feasible candidate sets already enforce the action grid, surge bounds, adjustment limits, and historical support requirements.

E.3 Solver and Independent Validation

HiGHS solves the MILP. A proven optimal solution or a feasible incumbent returned at the time limit may be accepted only if an independent validator recomputes network coverage, candidate eligibility, Booking Rate, nominal modeled GMV, and the objective from the selected candidates. This independent check prevents a solver status alone from being treated as evidence that the resulting network policy is valid. If no feasible solution exists or validation fails, GGSO produces no recommendation and the current prices remain unchanged.

F. Model Assumptions and Limitations

DuoCast assumes that historical temporal, spatial, and mobility patterns remain informative for the next five-minute interval. Forecast reliability may decrease when marketplace conditions differ substantially from the training data.

The response models are estimated from observational data. Their predictive performance does not establish that the estimated price effects are causal.

Surge pricing can induce spatial supply responses [2]. GGSO candidate simulation, however, holds capacity, fare, and pre-decision context fixed across price candidates and does not model driver redistribution. Cell-level conservative scores do not form a joint network-level confidence bound, and modeled no-loss does not guarantee realized no-loss.

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